# DevOps Lab Report

## Week – 4

| Group Members | USN |
|---|---|
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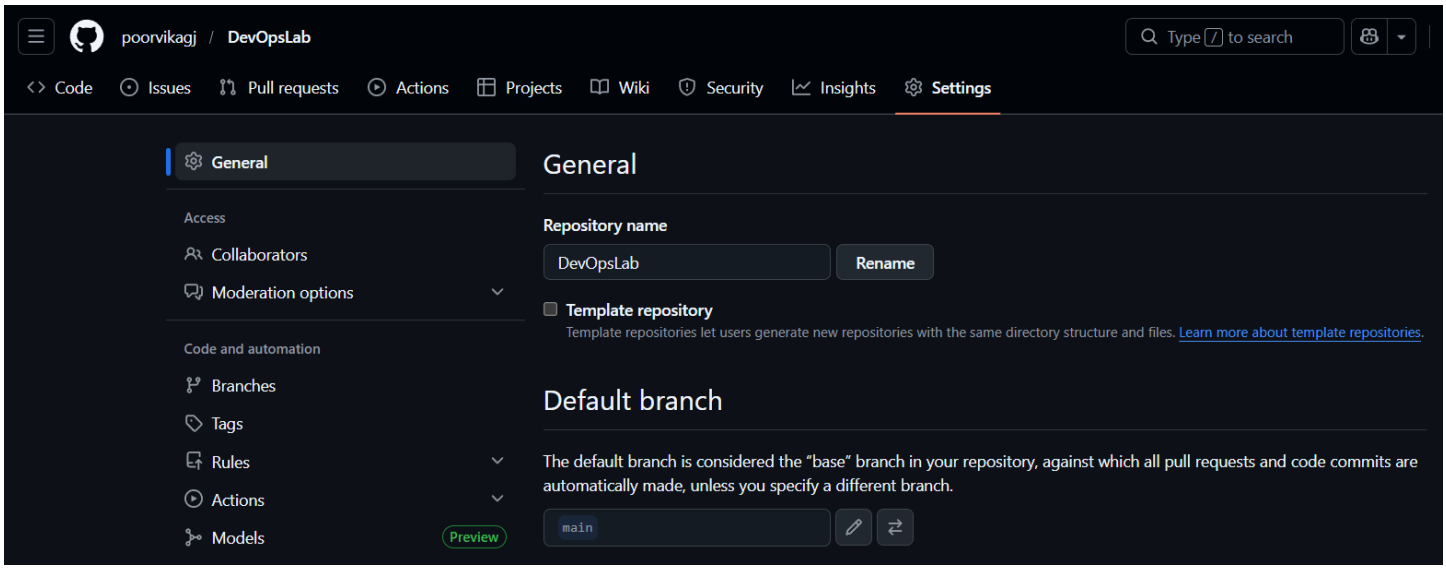
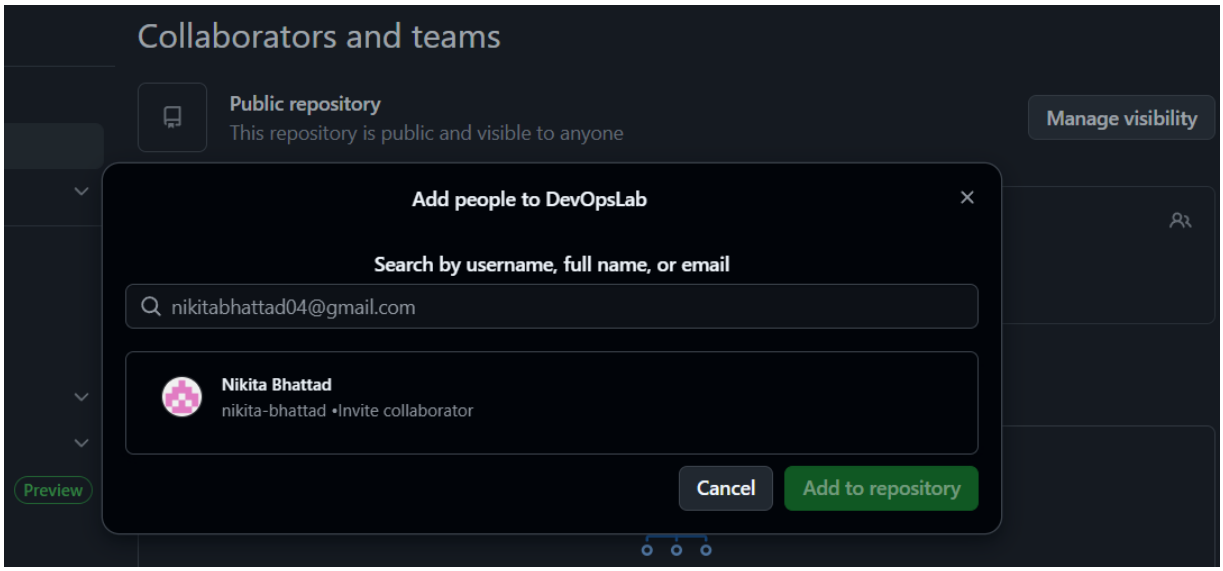
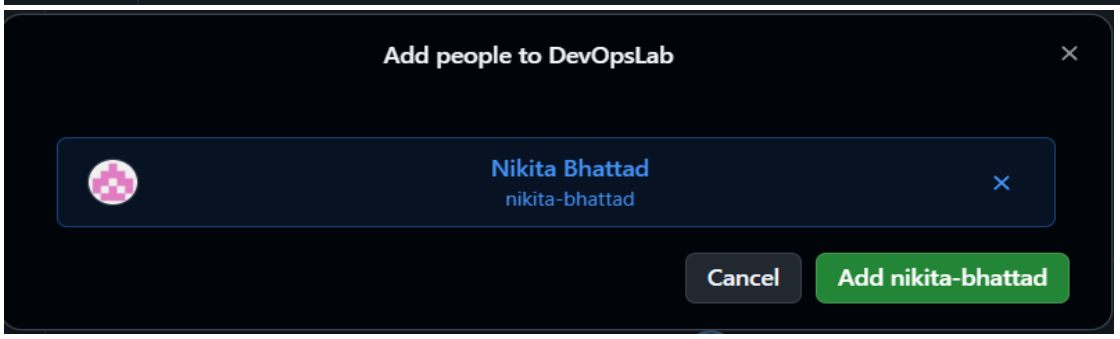
## Git collaborator

A **Git collaborator** is a person who is given permission to contribute to a repository in **Git** hosted on platforms like **GitHub**, **GitLab**, or **Bitbucket**.

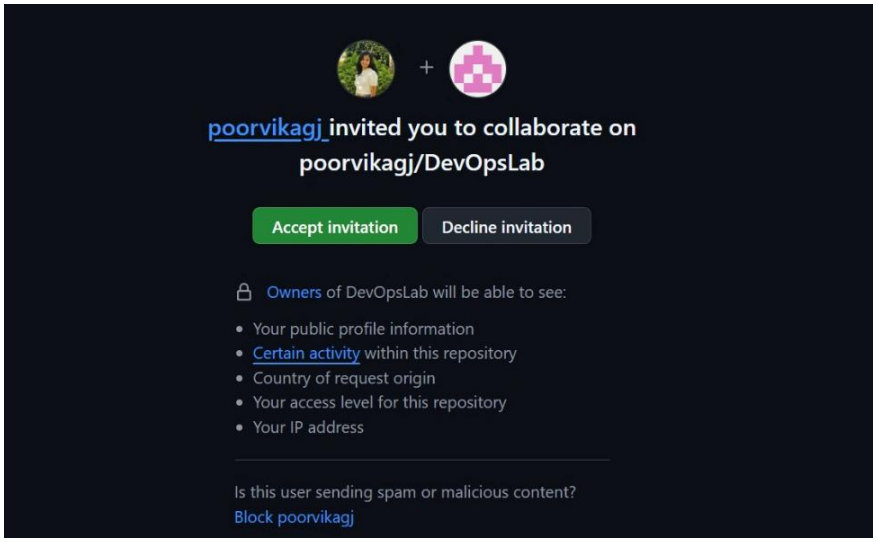
A **collaborator** is a user who has been granted access by the repository owner to **view, modify, and contribute to the repository's code and resources**.

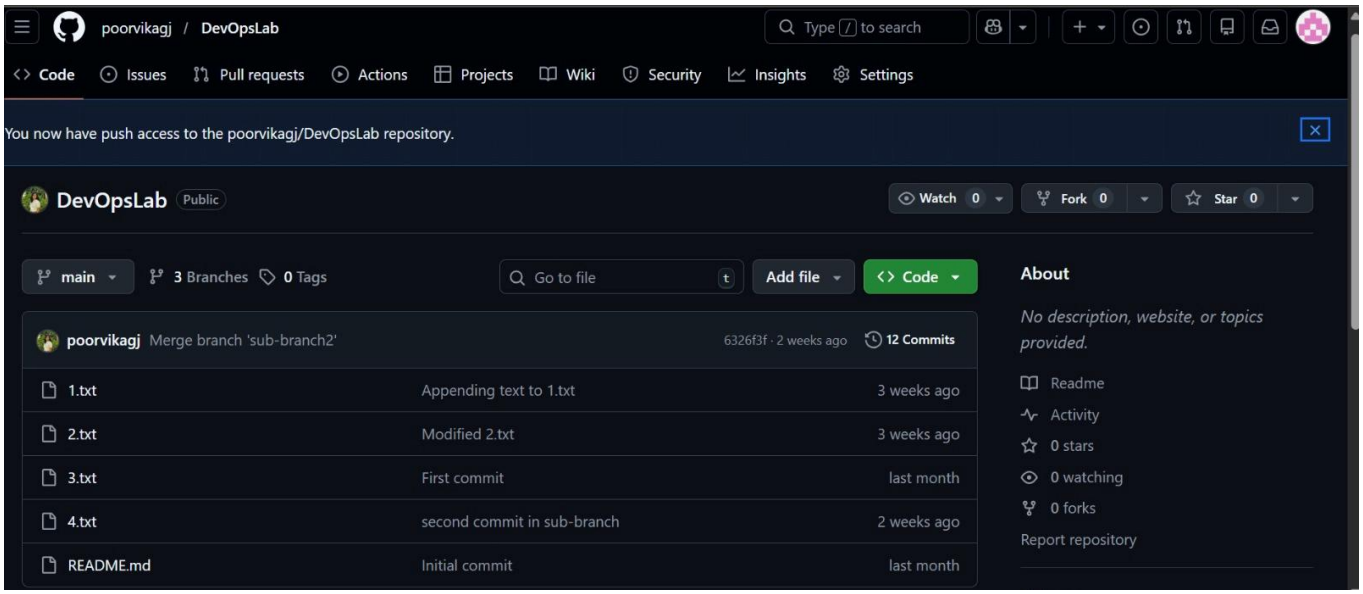


## The other User Side




## Git another remote repo

```
HP@DESKTOP-0GSVFEE MINGW64 ~/devOpsLab (branch1)
$ git remote add repo https://github.com/poorvikagj/dummy.git

HP@DESKTOP-0GSVFEE MINGW64 ~/devOpsLab (branch1)
$ git remote -v
origin  https://github.com/poorvikagj/DevOpsLab.git (fetch)
origin  https://github.com/poorvikagj/DevOpsLab.git (push)
repo    https://github.com/poorvikagj/dummy.git (fetch)
repo    https://github.com/poorvikagj/dummy.git (push)
```

## Installation of JDK (Java Development Kit)

**1. Download JDK 25**

1. Open a browser.
2. Go to the official Oracle JDK page:
   https://www.oracle.com/java/technologies/downloads/
3. Scroll to **Java SE Development Kit 25**.
4. Click **Windows**.
5. Download the file:
   **jdk-25_windows-x64_bin.exe**

**2. Install JDK 25**

1. Go to your **Downloads** folder.
2. Double-click **jdk-25_windows-x64_bin.exe**.
3. The installer will open.
4. Click **Next**.
5. Choose installation location (default is recommended): `C:\Program Files\Java\jdk-25`
6. Click **Next** again.
7. Wait for installation.
8. Click **Close** when finished.

**3. Set JAVA_HOME Environment Variable**

1. Press **Windows + S**
2. Search **Environment Variables**
3. Click **Edit the system environment variables**
4. In the **System Properties** window:
   - Click **Environment Variables**
5. Under **System Variables**
   - Click **New**
6. Add:

   **Variable Name:** JAVA_HOME

   **Variable Value:** `C:\Program Files\Java\jdk-25`
7. Click **OK**

**4. Add Java to PATH**

1. In **System Variables**, find **Path**
2. Select it and click **Edit**
3. Click **New**
4. Add: `%JAVA_HOME%\bin`
5. Click **OK → OK → OK**

**5. Verify Installation**

1. Open **Command Prompt**

   Press: Windows + R

   Type: `cmd`
2. Run: `java -version`

   You should see something like:

   java version "25"
3. Check compiler: `javac -version`

   Output:

   javac 25

## Installation of Jenkins

**1. Install Java (Prerequisite)**

Jenkins requires Java to run.

1. Open **Command Prompt**.
2. Check if Java is installed by running: `java -version`
3. If Java is not installed, install **JDK 17 or later**.

   (For example: **Java Development Kit 25**)

**2. Download Jenkins**

1. Open your browser.
2. Go to the official Jenkins website: https://www.jenkins.io/download/
3. Under **Long-Term Support (LTS)**, select **Windows**.
4. Download the installer file: jenkins.msi

**3. Run the Jenkins Installer**

1. Go to your **Downloads** folder.
2. Double-click **jenkins.msi**.
3. The **Jenkins Setup Wizard** will open.

**Installation Steps**

1. Click **Next**.
2. Choose the installation directory (default recommended): `C:\Program Files\Jenkins`
3. Click **Next**.

**4. Service Logon Credentials**

During installation you will be asked to choose how the Jenkins service runs.

**Option 1: Run service as a specific user (recommended for production)**

- Allows Jenkins to access network resources and shared drives.

**Option 2: Run as LocalSystem (default)**

- The **LocalSystem account** is a built-in Windows account with very high local privileges.
- It can start services and access local resources.
- However, it **cannot access domain resources or network shares that require specific user credentials**.
- Therefore, it is **not recommended for production environments**.

Choose the appropriate option and click **Next**.

**5. Configure Port Number**

Jenkins runs on a specific port.

1. The default port is: 8080
2. It is recommended to keep the default port.
3. Click **Next**.

**6. Choose Java Home**

The installer will ask for the **Java installation path**.

Example: `C:\Program Files\Java\jdk-25`

Select the installed JDK folder and click **Next**.

**7. Custom Setup**

1. Select any additional services that need to be installed with Jenkins.
2. Enable **Firewall settings** if prompted.
3. Click **Next**.

**8. Install Jenkins**

1. Click **Install**.
2. Wait for the installation to complete.
3. Click **Finish**.

Jenkins will start automatically as a **Windows service**.

**9. Open Jenkins in Browser**

Open your browser and go to: `http://localhost:8080`

You will see the **Unlock Jenkins** page.

**10. Get the Initial Admin Password**

1. Open **File Explorer**.
2. Navigate to: `C:\ProgramData\.jenkins\secrets`
3. Open the file: initialAdminPassword
4. Copy the password.
5. Paste it into the **Unlock Jenkins** page in the browser.

**11. Install Recommended Plugins**

After unlocking Jenkins:

1. Click **Install suggested plugins**.
2. Jenkins will download and install the required plugins.
3. Wait for the installation to finish.

**12. Create Admin User**

Fill in the following details:

- Username
- Password
- Full Name
- Email Address

Click **Save and Continue**.

**13. Start Using Jenkins**

Click:
Start using Jenkins
The **Jenkins Dashboard** will open.
You can now **create your first project or CI/CD pipeline**.